

Analyzing herding, stylized facts, and information cascades via self-organized criticality in an agent-based speculation game[☆]

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ARTICLE INFO

Keywords:

Agent-Based Model (ABM)
Behavioral finance
Herding
Information cascade
Self-organized criticality

ABSTRACT

This study advances Kai Katahira's Speculation Game, an agent-based model (ABM) for financial markets, by addressing its limitation in capturing order flow imbalance, a critical indicator of herd behavior. Although the original model successfully replicated key stylized facts of financial markets, it did not account for the persistence of order imbalance observed in real-world trading. Through a comprehensive analysis of two decades of BSE Sensex data, we establish the prevalence of order imbalance and its correlation with price fluctuations. To bridge this gap, we propose an extended model, Speculation Game with Information Cascade (SGIC), which integrates Self-Organized Criticality (SOC) through a sand-pile model, enabling agents to interact within a small-world network. Our proposed model not only reproduces the stylized facts captured by the original Speculation Game, but also successfully generates the additional stylized fact of order flow imbalance. These advances enhance the realism of ABMs in financial markets, providing deeper insights into the mechanisms driving herding and market fluctuations.

1. Introduction

The concept of SOC was introduced by Bak, Tang, and Wiesenfeld in 1987 [1] to describe how complex systems inherently develop into a critical state where minor events can have significant consequences. SOC is characterized by the following features:

- i Critical State: The system reaches a critical state, naturally through its internal dynamics without needing external fine-tuning, in which it is on the verge of a state change between order and disorder, where a small disturbance can escalate into a large event.
- ii Scale Invariance: This means that the statistical characteristics of the system remain consistent, regardless of the scale at which they are observed.
- iii Avalanches: In self-organized critical systems, small disturbances can lead to cascading effects, causing avalanches. These events can vary widely in size, but their occurrence follows a power-law distribution, a probability distribution characterized by the property that there are many small events and fewer large ones, represented as $P(X \geq x) \propto x^{-\alpha}$ where α is the scaling exponent that characterizes the power-law behavior.

[☆] This document is the product of research conducted at IIIT, Hyderabad, India.

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SOC provides a framework for understanding the dynamics of information diffusion and market fluctuations in financial markets [2] and suggests that the system is perpetually on the brink of a critical state, where minor perturbations can trigger widespread reactions. This is particularly relevant in the presence of information asymmetry, where some agents have access to more information than others. These asymmetries can result in cascading effects and rapid dissemination of information, consequential to significant market movements. Economists have long attempted to understand the instability of economic aggregates, but the tools and models currently used in economics have had limited success in managing and controlling these fluctuations [3]. The concept of Rational Expectations [4], which proposes that agents make predictions about future variables based on all available information, has not contributed to comprehending the overall financial behavior and the underlying causes of significant financial crises. Since 1970, the field of behavioral finance has captured growing attention and continued to evolve, with researchers exploring various aspects of human behavior and decision-making in financial markets. Researchers are exploring the constraints of the Efficient Market Hypothesis (EMH) [5], which proposes that financial markets efficiently integrate and reflect all important information. It has been observed by several researchers [6–8] that the behavior shown by investors does not follow the rational behavior assumed by traditional EMH. In the literature of modern finance, market agents are typically depicted as rational individuals with similar preferences, strategies, and expectations; however, Shiller [7] contends that psychological biases can significantly influence stock prices.

Since the seminal work of Arthur [9], researchers have explored alternative approaches to examine the behavior of stock prices, assuming that agents with bounded rationality inductively make decisions [10] and tend to simplify the complex task of making decisions by relying on a set of limited ‘heuristics’ [11,12]. Cognitive psychologists have extensively studied behavioral patterns such as availability bias, recency effect, base rate neglect, sample size neglect, conservatism bias, gambler fallacy, etc. [13]. Cont et al. [14] used mathematical models and simulations to explore herd behavior resulting in fluctuations in market prices and propose that herd behavior can amplify market volatility. Yaseen and Rao [15] analyzed daily stock data from the SSE and SZSE between 1992 and 2015 using a variety of information asymmetry metrics and found that there is a correlation between information asymmetry and a higher tendency of investors to exhibit herding behavior. The study by Tiniça, Iqbal, and Mahmud [16] gives significant insights into the intricate interactions between information cascades, short-selling constraints, and herding behavior in stock markets. It reveals that short-selling constraints can increase herding behavior, and the findings are consistent with those recorded by Yaseen and Rao [15]. In light of this, it has become crucial for market stakeholders to understand the factors behind herding behavior, its impacts, and to address challenges such as contagion effects and speculative activities. Fluctuations in economic aggregates can occur due to numerous minor disruptions in various sectors of the economy without the necessity of an external catastrophic force to cause significant adverse events and can be explained using SOC. In financial market ABMs, SOC can be used to capture the evolution of complex behaviors from basic interactions between traders or institutions.

2. Agent-based models

Researchers have developed various computer-simulated artificial financial markets consisting of agents capable of learning and optimizing to examine the decision-making processes of agents, their behavior, market rules and abnormalities in market dynamics. A review of the literature on ABMs in financial markets reveals a substantial body of research exploring the dynamics of complex financial systems. For instance, research by Cont [17] demonstrated that interactions among heterogeneous traders with different strategies can produce realistic price patterns and volatility clustering. ABMs have also uncovered the occurrence of herding behavior and information cascades in financial markets. Bikhchandani [18] showed that under certain conditions, even a minority of informed agents can influence the entire market, leading to herding and price bubbles. In addition, ABMs have been employed to explore market efficiency and the persistence of anomalies. Chiarella [19] found that incorporating behavioral components, such as noisy traders, can generate stylized facts and deviations from the Efficient Market Hypothesis (EMH). ABMs have further been used to study systemic risk and contagion in financial networks. Haldane and May [20] showed that interconnectedness and feedback loops between institutions can lead to cascading failures and systemic risk. Hommes [21] demonstrated that heterogeneous beliefs and trading strategies can result in complex market dynamics, including boom-and-bust cycles. In addition, ABMs have been utilized to explore the impact of market regulations and policy interventions. LeBaron and Tesfatsion [22] found that imposing circuit breakers can reduce market volatility and improve stability.

Heterogeneous Agent Models (HAM) assume that agents differ in their characteristics, strategies, and beliefs. The model by Bertella et al. [23] categorizes agents into fundamentalist and chartist groups, demonstrating that behavioral heterogeneity contributes to the excess volatility of risk assets relative to their underlying value. The Contagion Financial Pricing (CFP) model by Biondo et al. [24] investigates the impact of different agent types and information diffusion on market dynamics using the SOC framework, successfully replicating key stylized facts observed in financial markets.

2.1. Minority game-based models

In agent-based financial modeling, researchers strive to tackle the challenge of analyzing real market systems. These systems involve agents with similar capabilities fighting for limited resources. In this complex framework, each agent is knowledgeable about historical market data and must formulate trading decisions based on this shared global information. Among the different ABMs, the Minority Game (MG) [25] is particularly notable in that agents compete to be on the minority side. The MG model, inspired by the El Farol bar problem [9], requires agents to base their buy or sell decisions solely on “memory” of past outcomes, rewarding those who choose the minority option with a positive score while penalizing others. As agents continuously adapt their

strategies, the Minority Game reveals complex dynamics characterized by alternating periods of equilibrium and volatility. This adaptive behavior can lead to phase transitions within the system, resulting in patterns and behaviors that closely resemble those observed in real-world markets. The Complex Network Minority Game Model, proposed by Lingyun Chen [26], incorporates the dynamics of interconnected networks and the behavior of minority groups in decision-making processes within the market. It aims to capture the complexities and interactions present in real-world financial systems, providing insight into market behavior and dynamics through simulation.

In the speculation game, the decision-making structure of the Minority Game was extended by Kei Katahira [27]. Their model exhibits three distinct features when compared to previous ABMs:

- i It enables idling periods and non-uniform holding.
- ii It includes information on the magnitude of price changes in historical data.
- iii It implements a cognitive world to evaluate investment strategies, accounting for both capital gains and losses.

2.2. Agent-based models of behavioral dynamics

A variety of novel models were presented to explore the human influence on financial decision-making [28]. Shiller's work [7] ignited debate and led to the development of behavioral finance. The phenomenon of the information cascade and herding has a significant impact on financial market decision-making processes. An information cascade refers to a phenomenon in which individuals make decisions based on observing the actions of others rather than relying on their private information. Information cascades often result in herding, defined as the phenomenon wherein agents imitate group action rather than making independent decisions using their private information [29]. Numerous studies have empirically validated the presence of information cascades in financial markets. Bikhchandani [18] introduced the pioneering concept of information cascades, illustrating how sequential decision-making—where each participant is influenced by the actions of those before them—can result in the widespread adoption of particular investment choices. This herd behavior often arises from the desire to conform, mitigate potential losses, or seek social validation.

The “Self-Organizing Ising model of agents” [30], based on the principles of Ising model in physics, where each agent is viewed as a spin that has the state and interacts with its neighbors, demonstrates that by incorporating herding behavior, the impact of external news, and private information, the model successfully replicates the essential patterns in financial markets. When news about returns is true, agents may begin to assume that following the same information as others is a winning strategy. This dynamic pushes the system to function near the Ising critical point, where little adjustments can cause large market volatility. Boğaçhan [31] studied how agents make decisions and update their beliefs when they have access to the choices and outcomes of others but lack complete information about the environment. They concluded that observational learning can be effective in situations with imperfect information.

3. Problem statement

Minority Game-based models provide a powerful framework to simulate and understand financial market complexities by revealing how rational agents with limited information make decisions that collectively drive market dynamics. The Speculation Game (SG) model, rooted in the Minority Game framework, successfully replicates various stylized facts but notably lacks herding behavior, as evidenced by the balanced buy and sell orders. Multiple researchers have confirmed the presence of order imbalance which is further demonstrated in Fig. 1, which analyzes data from the BSE Sensex spanning December 2004 to February 2024. While U.S. indices like the S&P 500 or NASDAQ are commonly used, the BSE Sensex offers an opportunity to examine whether similar patterns hold in other markets, adding depth and diversity to global financial research. The BSE Sensex provides a rich and diverse dataset, capturing various economic cycles, financial crises, and recovery phases in the Indian market. This time span includes critical events such as the 2008 global financial crisis and the COVID-19 pandemic, making the data relevant for studying order imbalances.

The research aims to develop an Agent-Based Model (ABM) grounded in a Minority Game framework, enriched with behavioral influences, to simulate financial market dynamics and analyze herding behavior driven by information asymmetry. The study explores the use of SOC to model the spread of information and examines the integration of herding behavior within the ABM framework. The study uses stylized facts from financial time series to validate the ABM's performance, ensuring that the modeled herding behavior accurately aligns with observed market dynamics.

The model includes an interconnected network of agents that can be utilized to disseminate information and assist decision-making affected by herding. As empirical evidence from numerous studies [32] supports (i) the presence of information asymmetry and (ii) the existence of information cascades and herding behavior in financial markets, recognizing these phenomena is critical for understanding the market dynamics, the formation of asset prices, and the potential challenges they pose to market efficiency.



Fig. 1. Buy-Sell order imbalance represented as $H = \frac{\text{BuyOrders} - \text{SellOrders}}{\text{BuyOrders} + \text{SellOrders}}$ in the data of BSE Sensex from December 2004 to February 2024.

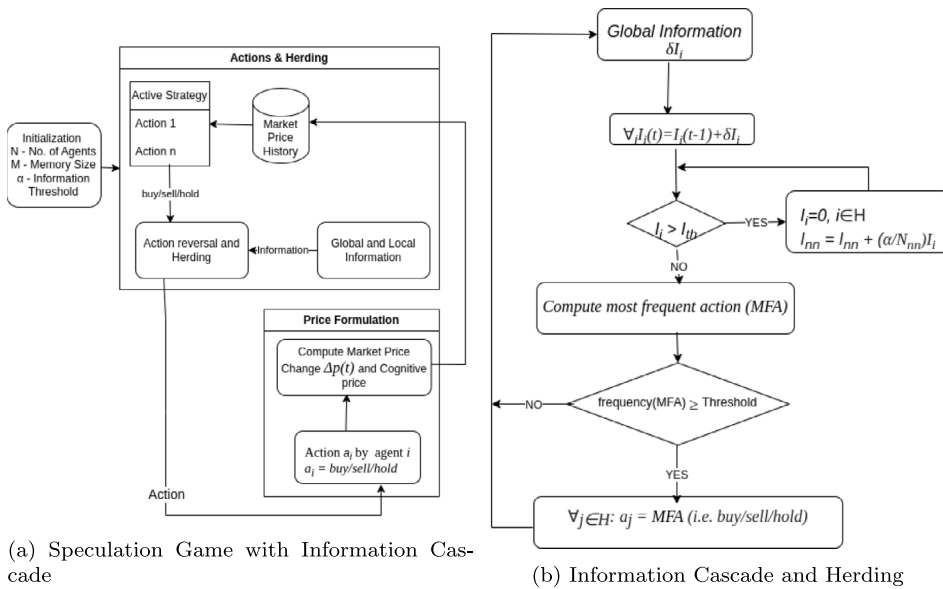


Fig. 2. Conceptual framework illustrating the emergence of herding behavior through information cascades.

Proposed model – speculation game with information cascade (SGIC)

The proposed model, SGIC, illustrated in Fig. 2(a), extends the Speculation Game (SG) model, incorporates information cascades, and herding behavior to reflect real-world dynamics better. The implementation of the model is described below.

1. N agents are connected in a Small World (SW) network to receive and share information and compete to enhance their wealth through capital gains via round-trip trades. An SW network has been chosen over a scale-free network due to two key properties it exhibits: high clustering and short average path lengths. This clustering reflects the presence of tightly-knit communities within the overall network, as herding emerges from local information-sharing rather than centralized hubs dominating the flow as in case of scale-free networks. Despite this clustering, any two nodes in the SW network are not far apart, allowing them to communicate quickly and efficiently even if they are not directly connected.
2. At each passing time step t , the agent i takes an action:

$$a_i(t) \in \{1, -1, 0\}$$

i.e., buy (1), sell (−1), or hold (0), based on the history representing the patterns of the previous price movements.

3. Agents continuously receive global as well as local information that may alter the action a_i of the agent i . The global information reflects the overall atmosphere of the market that each agent senses through news, such as economic indicators, geopolitical events, corporate earnings reports, etc. The local information is received from peers, and people pass information

to peers for several reasons, grounded in psychological, social, and cognitive theories such as social proof and validation, cognitive relief, influence and persuasion etc [33,34].

4. $I_i(t+1) = I_i(t) + \delta I_i$ where $I_i(t)$ is the information possessed by agent i , $\delta I_i \sim \text{Uniform}(0, I_{th} - I_{\max}(t))$, $I_{\max}(t) = \max(\{I_i(t) \mid \forall i\})$ and I_{th} is a threshold value that is same for all agents.
5. Agents decisions may be affected by the actions of well-informed peers, i.e., information asymmetry, leading to a cascade effect where others follow the actions of these influential agents, and the same is implemented using the concept of SOC. When the information with agent k becomes greater than their personal threshold value I_{th} , k becomes “active” and transmits the informative signal, an individual stream, to its neighbors according to the following mechanism [24]:

$$I_k > I_{th} \Rightarrow \begin{cases} I_k \leftarrow 0, \\ I_{nn} \leftarrow I_{nn} + \frac{\beta}{N_{nn}} I_k, \end{cases} \quad (1)$$

N_{nn} represents the number of direct neighbors of the active agent k , and the β is used to decide how much information is to be dissipated during the experiment. As agents receive the information from their neighbors, some of them may also become active, as $I_i(t) > I_{th}$, and in turn transmit their signal to their neighbors, resulting in an informative avalanche that may sometimes be very high, involving a large number of active agents. The flowchart of this implementation is shown in Fig. 2(b). This implementation of information avalanche is based on the Sand Pile model of SOC, where:

- i $I_i(t)$ is the sand-pile height.
- ii I_{th} is the critical height at which an avalanche occurs.
- iii $I_i(t+1) = I_i(t) + \delta I_i$ represents the addition of sand grains to the sand pile. In the model, the agents are conceptualized as individual cells within a two-dimensional cellular automaton in the Abelian sand-pile model [35]. Each cell i.e. agent accumulates discrete units of information over time, analogous to sand grains in the sand-pile model. However, unlike in the Abelian sand-pile model where each grain is identical, the information received by the agents varies in significance and impact. This reflects the diverse nature of information flow in the market, where different agents experience varying levels of influence based on the specific information they receive.
- iv $I_{nn} \leftarrow I_{nn} + \frac{\beta}{N_{nn}} I_k$ represents the redistribution of sand grains i.e. the local information, from an agent to its neighboring agents.

The implementation of SOC in Self Organized Speculation Game (SOSG) model [36] and SGIC model differs fundamentally in the way critical states emerge and propagate within their respective financial market frameworks. In the SOSG model, the gamified market represents the lattice, and new players entering the market resemble grains of sand being added to the system. Each player enters the market with an initial wealth of $w_i(t) = 10B$, where B is the board lot amount that determines the ease of placing orders in multiple quantities. If a player's wealth drops below B , they are forced to exit the market, leading to irregular withdrawals that mimic the toppling of sand grains when a critical slope is reached in a sand-pile. Thus, while both models exhibit self-organized criticality in financial markets, the SOSG model attributes criticality to player exits driven by a wealth threshold, whereas the SGIC model links it to information cascades that influence trading decisions.

6. To implement herding behavior, the model must have the following provisions:

- i to specify the decision-making rules for the agents,
- ii a mechanism for herding so that agents can observe the actions of their neighbors and adjust their decisions accordingly.
- iii a herding threshold parameter, θ , that determines the proportion of other agents an individual agent needs to observe before deciding to follow the crowd.

All those agents who become active by crossing their threshold value as per the above-mentioned process will form a herding group H . To implement herding, the herding decision rule is to be identified, which defines how agents decide to imitate the actions of other agents. The decision rule plays a crucial role in determining when agents decide to align their actions with those of their peers. In the experiment:

$$A_H = \begin{cases} \arg \max_{a \in \{Buy, Sell, Hold\}} \sum_{i \in H} \mathbb{1}(a_i = a) & \text{if } \max_a \sum_{i \in H} \mathbb{1}(a_i = a) \geq \theta \\ No_Change & \text{otherwise} \end{cases} \quad (2)$$

Here

- (a) The equation states that if the frequency of most frequent action meets or exceeds θ , then all agents will follow that action i.e. as per Bikhchandani et al. model [37], investor's decisions will be reversed after observing the actions of others, exhibiting herding behavior: $\forall i \in H, a_i = A_H$; otherwise no change occurs in the agents' actions.
- (b) θ represents the threshold required to trigger herding behavior, which is set at 40% in the experiments based on the concept of Majority Rule in Herding. This choice accounts for the fact that agents in the model can take one of three possible actions: Buy, Sell, or Hold. In traditional binary decision-making systems, a simple majority threshold (more than 50%) is typically required to drive consensus. However, in a three-choice scenario, the influence of any single action must be assessed differently. Since there are three options, a strict majority (above 50%) would be too

Table 1
Model parameters summary.

Parameter	Value
Number of agents (N)	1000
Memory Size (M)	5
Board Lot (B)	9
Cognitive Threshold (C)	3
Number of Strategies (S)	1
Neighbors of an agent (k)	40
Information Threshold	1.0
Information Dissipation β	0.98

Table 2
ADF test summary.

Parameter	Test statistic	P
Herding	-19.5948	0.00
Volatility	10.9891	0.00
Returns	-42.2768	0.00

restrictive, allowing herding to emerge only in extreme cases. Instead, a lower threshold, 40%, ensures that one action is significantly more dominant than either of the other two while still allowing for meaningful group behavior without requiring an absolute majority.

- The market price dynamic Δp will be calculated afterward as per the Speculation Game based on the assumption that return Δp is proportional to excess demand [14].

$$\Delta p = \sum_{i=1}^N a_i(t)q_i(t)/N$$

4. Simulations, observational results, and contextual discussion

During the experimental investigations, both the Speculation Game (SG) Model and the proposed SGIC model were subjected to extensive simulations. The results were carefully analyzed and then compared. To ensure consistency, the experimental conditions were standardized with fixed parameter values, as outlined in Table 1.

4.1. Model validation

To make reliable inferences, the model has been validated for stationarity using the Augmented Dickey–Fuller (ADF) unit root test [38] and by assessing whether the simulated data generated by the model exhibits key empirical regularities in real-world data.

4.1.1. Stationarity

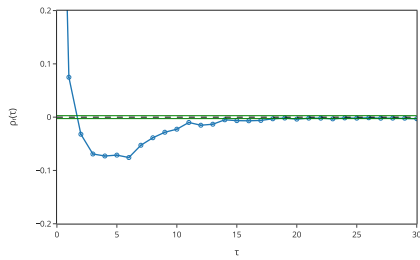
Stationary time series are characterized by constant statistical properties over time, making it easier to model and analyze them. ADF test results contribute to the validation of time series models. By confirming or rejecting the presence of a unit root, researchers can make informed decisions about the appropriateness of the models for analysis. The ADF test is conducted by using all parameters slated for exploration, encompassing aspects such as market price, return, absolute returns, and herding. Table 2 shows the results of ADF test. Non-stationarity is the null hypothesis. The statistics generated by the test and the corresponding p-values were analyzed, resulting in the rejection of the null hypothesis.

4.1.2. SGIC model validation using stylized facts

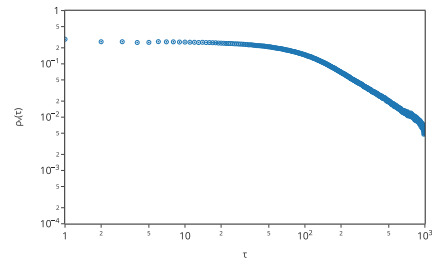
Stylized facts are recurring patterns commonly observed in financial time series that provide a benchmark for validating the accuracy of an ABM. Ensuring that the SGIC model aligns with these stylized facts helps confirm that it accurately represents key features of real-world financial systems. Cont [39] extensively discusses these stylized facts, and a similar analysis has been conducted by Jie-Jun Tseng [40]. To validate SGIC for stylized facts, the price return has been computed as $r(t) = \ln(p(t)) - \ln(p(t-1))$. The following stylized facts have been used for validation: intermittency, fast decay of autocorrelation, clustering of volatility, aggregated gaussianity, leverage effect, volume/volatility correlation, and asymmetry in time scale, heavy tails in the distribution of returns.

The auto-correlation in returns generated by SGIC, calculated as, $\rho_r(\tau) = \text{Corr}(r(t+\tau), r(t))$, where τ is time lag, decays rapidly, as shown in Fig. 3(a), that becomes nearly non-significant after 17 time lags, illustrating the absence of auto-correlation in returns. This outcome is the average of 100 simulation trials consisting of 50,000 time steps. Another stylized fact, volatility clustering computed as auto-correlation in absolute returns, $\rho_v(\tau) = \text{Corr}(|r(t+\tau)|, |r(t)|)$, diminishes at a slower rate, and the same has been indicated in the data generated by SGIC in Fig. 3(b).

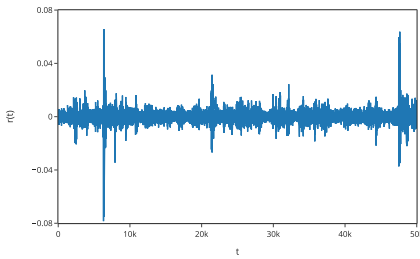
The intermittency is characterized by chaotic shifts between high and low volatility phases, with varying durations of these transitions. The time series of absolute return, $r(t)$ produced by SGIC presented in Fig. 3(c), shows the presence of intermittency.



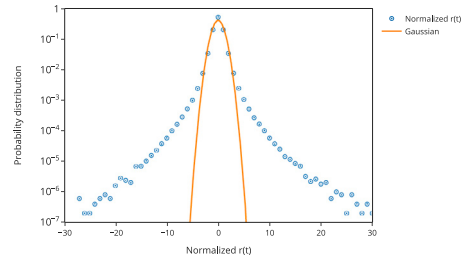
(a) Fast decay of linear autocorrelation in returns



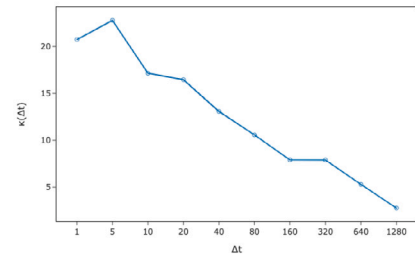
(b) The slow decay of autocorrelation of volatilities



(c) Intermittency in SGIC



(d) Presence of heavy tails, evidence of non-Gaussian behaviour.



(e) Recovering of the Aggregational Gaussinity in SGIC Model

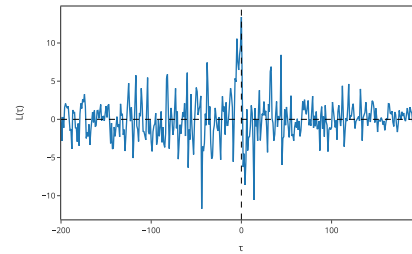
(f) $L(\tau)$ of SGIC Model

Fig. 3. Stylized market facts captured by the proposed model.

The heavy tail in time series indicates that the distribution of returns often has heavier tails than a normal distribution, meaning extremely large price movements occur more frequently. Fig. 3(d) shows the averaged probability distribution of $r(t)$ from the SGIC model, based on 100 trials of 50,000 time steps each. The returns are normalized and displayed on a logarithmic scale. The figure demonstrates that the SGIC model exhibits heavier tails in return distributions compared to the Gaussian distribution, as illustrated by the brown line.

Aggregational Gaussianity states that the distribution of returns becomes normal when we decrease the frequency of observation [41]. The Stylized Fact of Aggregational Gaussianity is closely tied to the Central Limit Theorem, which helps explain why some aggregate market variables display Gaussian properties. According to this theorem, the distribution of the sum of many independent and identically distributed random variables tends to approach a Gaussian form, regardless of the original distribution's shape. The excess kurtosis is a quantitative measure that can be employed to characterize the variation in distribution shapes compared to a Gaussian distribution. Kurtosis, $\kappa(\Delta t)$, large positive value indicates that the return distribution possesses a sharper peak and heavy tails. Consequently, it can be said that aggregational Gaussianity is a phenomenon where the $\kappa(\Delta t)$ diminishes with an increase in the time scale Δt . In Fig. 3(e), the alteration in the averaged kurtosis (over 100 trials) is illustrated, employing various time scales for calculating returns in SGIC. In Fig. 3(e), it is observed that $\kappa(\Delta t)$ decreases as the time scale increases, aligning with the notion that the return distribution exhibits a power-law tail beyond the Lévy stable regime.

Leverage effect refers to the tendency of asset prices to exhibit asymmetric responses to changes in the overall market or index level. It is observed that the volatility of stocks generally increases when the price decreases. Measuring the leverage effect, $L(\tau)$,

with time lag τ , involves computing correlation coefficients between asset returns and changes in volatility.

$$L(\tau) = \langle |r(t + \tau)|^2 r(t) \rangle / \langle r(t)^2 \rangle^2 \quad (3)$$

Negative values of τ lead to very small values of the correlation function, indistinguishable from noise, as evident from the figure. Analysis of the time-series produced by SGIC reveals the presence of leverage effect. Fig. 3(f) displays the averaged $L(\tau)$ and $\text{Corr}(|r(t + \tau)|^2, r(t))$ respectively for 100 trials each of 50,000 time steps. The figure depicts that for positive and small values of τ , there is a noticeable drop in the value of $L(\tau)$ and $\text{Corr}(|r(t + \tau)|^2, r(t))$. The correlation between volume and volatility refers to the observation that volumes of trading tend to increase when volatility is high in the market and vice versa. By permitting orders with variable quantities, the SGIC is capable of producing a correlation between volume and volatility. The buy and sell order quantities, $Q_{\text{buy}}(t)$ and $Q_{\text{sell}}(t)$, are computed as:

$$Q_{\text{buy}}(t) = \sum_{i=1}^N q_i(t) \delta(a_i(t), 1) \quad (4)$$

$$Q_{\text{sell}}(t) = \sum_{i=1}^N q_i(t) \delta(a_i(t), -1) \quad (5)$$

where the generalized Kronecker deltas are defined as follows:

$$\delta_{a_i}(t), 1 = \begin{cases} 0 & (a_i(t)) \neq 1, \\ 1 & (a_i(t)) = 1, \end{cases} \quad \delta_{a_i}(t), -1 = \begin{cases} 0 & (a_i(t)) \neq -1, \\ 1 & (a_i(t)) = -1, \end{cases} \quad (6)$$

$\langle V(t) \rangle$, the averaged trading volume, for time scale Δt , is calculated as follows:

$$\langle V(t) \rangle = \max \{ \langle Q_{\text{buy}}(t) \rangle, \langle Q_{\text{sell}}(t) \rangle \} \quad (7)$$

The $\langle \rangle$ is used to represent the average over the given time scale. The scatter diagram in Fig. 4(a) shows the strong correlation, $\text{Corr}(\langle V(t) \rangle, \langle |r(t)| \rangle)$, between the volume and volatility for $\Delta t = 5$ in the time-series produced by SGIC.

Asymmetry in time scales pertains to volatilities at various time resolutions. It is observed that coarse-grained volatility, $v^c(t)$, is more effective at predicting fine-scale volatility, $v^f(t)$, than vice versa. Coarse-grained volatility refers to a method of analyzing volatility that involves grouping data into broader time intervals. While fine-scale volatility refers to the analysis of price changes and fluctuations at a very detailed or granular level, often involving short time intervals, such as intraday data. For time scale Δt , $v^c(t)$ and $v^f(t)$ are computed as:

$$v^c(t_i) = \left| \sum_{j=1}^N r(t_{i-1} + j\Delta t, \Delta t) \right| \quad (8)$$

$$v^f(t_i) = \frac{1}{n} \sum_{j=1}^N |r(t_{i-1} + j\Delta t, \Delta t)| \quad (9)$$

Δt is $(t_i - t_{i-1})/n$. The formula to compute lead-lag correlation represented by $\rho_{cf}(\tau)$ between $v^c(t)$ and $v^f(t)$ is:

$$\rho_{cf}(\tau) = \text{Corr}(v^c(t + \tau), v^f(t)) \quad (10)$$

Fig. 4(b) illustrates $\rho_{cf}(\tau)$ alongside $v^c(t)$ and $v^f(t)$, computed at regular 50-time-step intervals with $\Delta t = 10$ and $n = 5$. The $\rho_{cf}(\tau)n$ is determined using averaged price returns of 100 trials each of 50,000 time steps. The linear asymmetry of $\rho_{cf}(\tau)$ concerning $\tau = 0$ is evident in the data produced by SGIC, as shown in Fig. 4(b). Moreover, the difference in correlations, $\rho_{cf}(\tau) - \rho_{cf}(-\tau)$ represented by the red color in the figure, is also not inside the 95 percent confidence interval of Brownian motion when $|\tau| \leq 3$.

5. Analyzing herding

In emerging markets such as India and China [42], individual investors often perceive the stock market as a gamble, driven by fear of missing out (FOMO) and a tendency to follow the herd in pursuit of quick profits. Herding behavior fueled by misinformation [43] or deepfake content, deliberately disseminated via social media, can artificially inflate market volatility, affecting both intraday stock prices and long-term valuations detached from fundamental market principles.

In the age of social media, the rise of market influencers and the premeditated spread of rumors—such as short-and-distort tactics—can reinforce confirmation bias, leading to large-scale speculative herding. This manipulation is further exacerbated by algorithmic trading, where automated strategies exploit herd behavior for profit.

The ethical implications of deliberately inducing herd behavior through information manipulation are profound. Intentionally misleading investors not only distorts market efficiency but also erodes public trust, disproportionately harming retail investors who rely on accurate market signals. If left unchecked, such practices can lead to financial instability, wealth inequality, and systemic economic crises.

The GameStop Short Squeeze of 2021 serves as a reminder of the disruptive power of social media in financial markets. In this context, alert and efficient regulatory bodies must take on an expanded role, ensuring the swift removal of misleading content to safeguard market integrity and protect investors from ethically questionable trading practices.

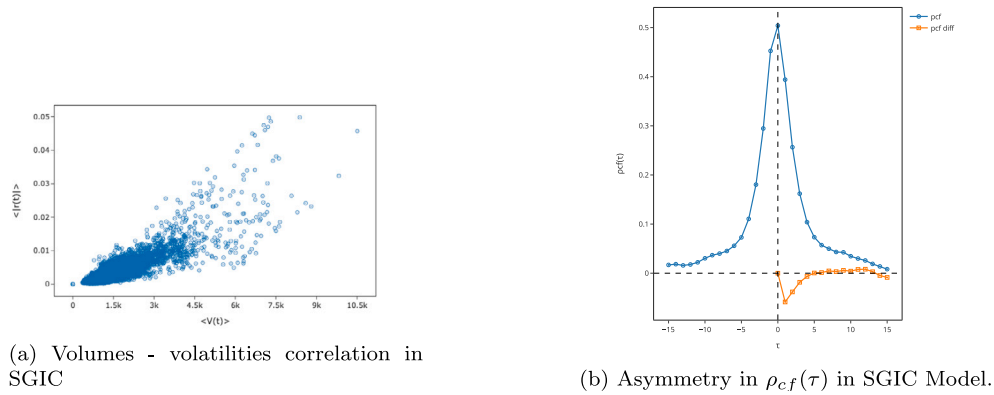


Fig. 4. Stylized facts captured by SGIC related to volatility dynamics and temporal asymmetry.

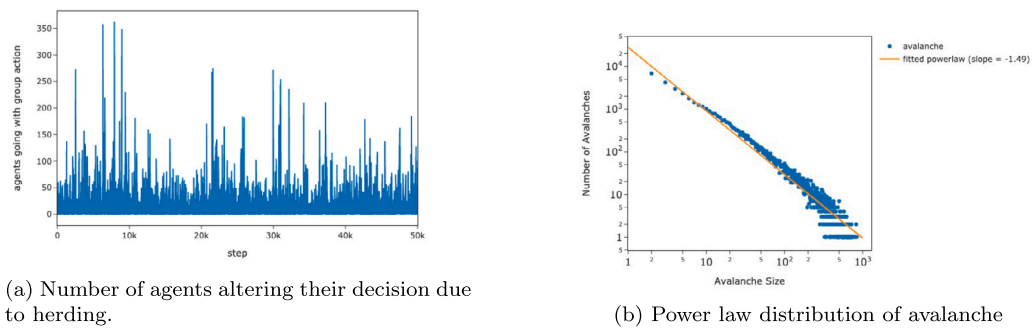


Fig. 5.

Fig. 5(a) illustrates how information cascades and herding behavior drive decision-making in the SGIC model, highlighting how individuals adjust their choices based on peer influence and available information, with the graph depicting the number of agents who altered their decisions in response to these dynamics. In systems dominated by herding behavior, individuals heavily influence each other's actions, generating feedback loops that amplify small disturbances and push the system toward critical states, risking instability. Minor shifts in investor sentiment can lead to large-scale fluctuations in asset prices. Fig. 5(b) displays the avalanche size against the number of avalanches generated during the SGIC model simulation, revealing a clear power-law distribution. A power-law distribution is identified by a straight line on a log-log plot, with the slope representing the distribution's exponent. In this case, the exponent (α) is 1.49, signaling the presence of SOC-like behavior in the system. This suggests that the model mirrors the dynamics of complex systems evolving toward a critical state, where power-law distributions dominate, and small events can trigger large-scale cascades, much like avalanches.

5.1. Interpreting order imbalance and herding in SGIC

Kei Katahira Speculation Game findings show the absence of herding as indicated by the stability of fluctuations in the quantities of agents submitting orders, which is in contrast with several well-documented findings [44]. Fig. 6 displays the correlation between the Herding Index (H) and the Percentage Index Variation (PIV), calculated using a 15-day window preceding the H date. The data, sourced from the BSE Sensex from December 2004 to February 2024, uses the H metric to assess the level of herding behavior among investors. Two primary factors are considered in this analysis:

1. Order Imbalance: This occurs when there is an excess of buy orders compared to sell orders or vice versa. A significant imbalance suggests that investors are disproportionately favoring one side, indicating potential herd behavior.
2. Index Variation: The measure of index fluctuation over a specific period. The high variation reflects market volatility, hinting that investors might be reacting emotionally rather than rationally.

A high H computed from an order imbalance on a particular day is a consequence of a strong herd behavior and, therefore should be preceded by volatile index variation. Correlation has been computed between the H and PIV in Preceding as well as

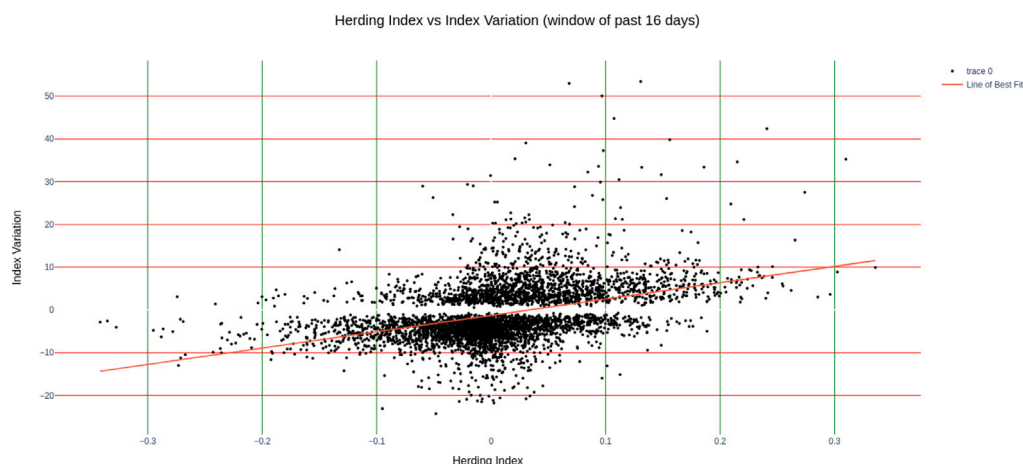


Fig. 6. H vs PIV from data of BSE Sensex from Dec, 2004 to Feb, 2024.

Table 3

Herding index vs percentage index variation.

Date	Herding Index	Index variation
13-Dec-2004	0.00131543	0.75868758
14-Dec-2004	-0.01174065	-1.45426549
15-Dec-2004	-0.03142241	-2.63577564
16-Dec-2004	-0.01731859	-2.91010812
17-Dec-2004	-0.01248388	-1.22524612

Succeeding windows of 2 to 30 days (Shown in Fig. 7 and Table 6 in the Appendix). PIV is computed as:

$$PIV = \begin{cases} \frac{LD}{CD} \times 100 & \text{if } |LD| > |HD| \\ \frac{HD}{CD} \times 100 & \text{otherwise} \end{cases} \quad (11)$$

where

- (a) X represents the current day
- (b) N represents the number of days in the window i.e. window size
- (c) d represents individual days within the window
- (d) Window type = Preceding or Succeeding
- (e) Preceding Window = $\{d | X - N \leq d \leq X - 1\}$
- (f) Succeeding Window = $\{d | X + 1 \leq d \leq X + N\}$
- (g) CD = Closing index on the current day
- (h) LD = Lowest closing index in the window - CD
- (i) HD = Highest closing index in the window - CD

Part of the table generated from December 2004 to February 2024 is given in Table 3. H is computed as: $H = (Total_buy - Total_Sale) / (Total_buy + Total_Sale)$. Where $Total_buy$ is the sum of buy orders of all categories of investors and $Total_sales$ is the sum of sales. The different categories of investors are Client, NRI, Proprietary, IFI, Banks, Insurance, and DII (Domestic Institutional Investors).

From Fig. 7, it is evident that the correlation between H and PIV in the preceding window remains consistently in the range of 0.3 to 0.4, suggesting a moderate backward-looking relationship—i.e., elevated herding levels tend to follow episodes of significant price movement. Conversely, the correlation in the succeeding window hovers close to zero, indicating that the current level of H does not provide meaningful predictive insight into forthcoming index variations. A correlation coefficient of approximately 0.4 between PIV of a preceding window of 16 days and the H suggests a moderately positive correlation between these two variables. Although it is not a perfect correlation, there is still a noticeable relationship between PIV and H . The correlation suggests that herding behavior may be influenced by recent price movements. Traders may be reacting to price changes by adjusting their buying or selling behavior, leading to observable patterns in the order flow.

As revealed from Table 4, Standard deviation and mean of H are nearly identical in the time series from BSE Sensex and that produced by SGIC. However, the standard deviation is approximately 10 times larger than the mean, indicating a significant degree of variability within the H data. Despite this variability in order imbalance, similar variability in price variation is not observed in either dataset. This suggests a recognizable relationship between price variation and herding behavior—volatile market conditions prompting changes in investor buy/sell behavior and herding. However, this variability in order imbalance may not necessarily result in comparable variability in price movements.

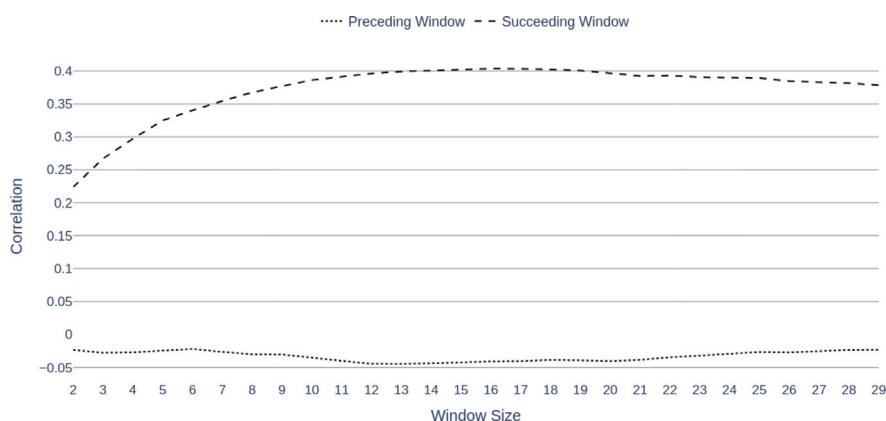


Fig. 7. Correlation ρ between H and PIV in the data from BSE Sensex from Dec 2024 to Feb 2024. Dashed line represents the ρ between H and PIV in preceding window and dotted line in succeeding window.

Table 4
Analysis of BSE and SGIC Data.

	SGIC	BSE
Mean of H	0	0.007
Standard deviation of H	0.074	0.072
Mean of price	104.127	0.3294
Standard deviation of Price	0.816	0.2443
correlation between price and H	0.128	0.162

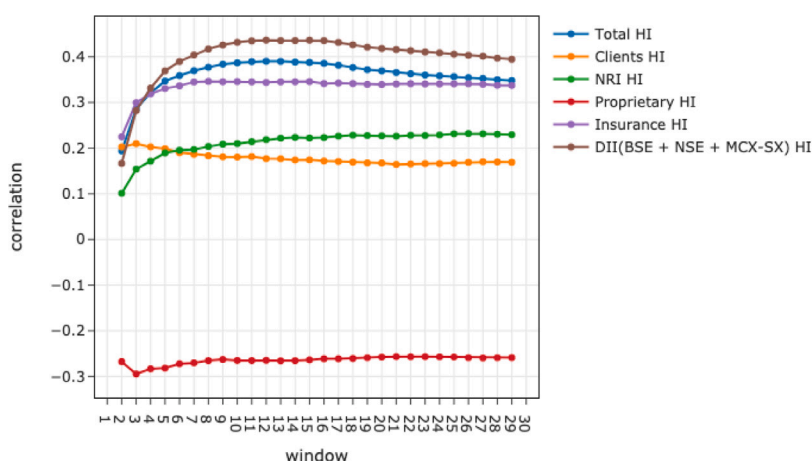


Fig. 8. Correlation ρ between H and PIV in the data from BSE Sensex from Dec 2024 to Feb 2024 when H is computed categorywise.

Fig. 8 illustrates the correlation between H and PIV across various window sizes using BSE data, with H calculated according to investor categories. It demonstrates that the buying and selling activities of proprietary investors tend to move in the opposite direction compared to those of other investor categories. Their buying activity during a bullish market and selling during bearish may reflect a belief that the upward or downward momentum will continue, while other investors may be more risk-averse or cautious. This helps in reducing bid-ask spreads.

The H exhibits certain limitations, and its interpretation necessitates careful consideration of the specific contextual application. In financial markets characterized by inherent noise, short-term fluctuations in trading patterns are susceptible to influence from diverse factors unrelated to herding behavior. Temporary spikes in trading volume can potentially generate misleading herding signals if not rigorously examined to discriminate between stochastic fluctuations and genuine herding dynamics. H is only the indicator of asymmetry in the number of buyers and sellers, which may not exclusively stem from herding behavior. Herding is defined as a deliberate effort by investors to emulate the actions of their counterparts. It is essential to differentiate this intentional herding from "spurious herding", where groups, confronted with comparable decision challenges and information, make similar choices. Unlike herding, spurious herding does not involve investors reversing their decisions after observing others. Instead, they might be responding to publicly available information that is widely known.

Table 5
Effect of K and θ on Power-Law exponent α .

K	θ	α	K	θ	α	K	θ	α	K	θ	α
20	40	1.48	40	40	1.47	60	40	1.46	80	40	1.45
20	50	1.48	40	50	1.47	60	50	1.46	80	50	1.45
20	60	1.48	40	60	1.47	60	60	1.46	80	60	1.45
20	70	1.48	40	70	1.47	60	70	1.46	80	70	1.45

5.2. Effect of different values of K and θ

The results from multiple experimental runs are tabulated in Table 5. The power-law exponent (α) characterizes the distribution of avalanche sizes, a key indicator of SOC. The general form of the power-law distribution in SOC is $P(s) \propto s^{-\alpha}$. It is evident from the table that by increasing θ (from 40 to 70) does not significantly change the power-law exponent. Since θ is a relative threshold, once a critical fraction of the population aligns, the herding process remains similar regardless of the small increase in θ .

As K increases from 20 to 80, the power-law exponent decreases slightly (from 1.48 to 1.45). More connections allow information to spread faster, leading to larger and more frequent herding events. This results in a slightly lower exponent, meaning large-scale events are more probable. The system is more globally connected, leading to more rapid cascades.

As evident from the discussion, a very low K could restrict the flow of information and delay the formation of avalanches, whereas a very high K may unrealistically approximate a fully connected network, where each agent has access to all others. Choosing $K = 40$, in the experiments ensures a sparse but well-connected topology, preserving the small-world characteristics of short average path length and high clustering.

The observed power-law exponent $\alpha \approx 1.45 - 1.48$ consistently emerged across multiple experimental runs and remained stable under different random seeds. This robustness suggests that the system inherently evolves toward a critical state, reinforcing the presence of Self-Organized Criticality in financial markets. The consistency of α further validates the model's ability to capture realistic market dynamics, where both small and large fluctuations coexist in a scale-free manner.

6. Conclusion

This study investigates the role of herding behavior in financial markets through an Agent-Based Modeling (ABM) approach, utilizing the Speculation Game with Information Cascade (SGIC) model. The analysis of BSE Sensex data reveals a strong correlation between market fluctuations and herding tendencies, quantified using the herding index derived from order imbalance—a key measure of collective investor behavior. Speculation Game (SG) based on the minority game, fail to capture herding behavior as indicated by nearly balanced buy and sell orders whereas order imbalance is highly persistent in the market [45]. However, the SGIC model successfully incorporates herding mechanisms through information asymmetry, cascades, and self-organized criticality (SOC), leading to a more comprehensive replication of real-world financial patterns.

6.1. Key contributions

1. Overcoming SG Model Limitations — The SGIC model extends the Speculation Game (SG) by introducing order imbalance, a persistent market feature previously absent in the SG model. This enhancement allows for a more realistic representation of investor behavior.
2. Integration of Behavioral Dynamics — The inclusion of information cascades and herding mechanisms in ABMs demonstrates that psychological and behavioral factors are critical in shaping financial market dynamics. These elements enhance the model's ability to reproduce empirical stylized facts.
3. Empirical Validation and Predictive Power — The SGIC model not only replicates key stylized facts observed in financial markets but also captures power-law distributed herding-induced avalanches, aligning with empirical data. This suggests that the modifications made to the model are robust and effective.
4. Implications for Financial Modeling — By refining the understanding of herding behavior, the SGIC model provides valuable insights for policymakers and financial analysts. It enhances predictive capabilities, making it a useful tool for risk assessment and market regulation strategies.

6.2. Future directions

There is a fundamental distinction between intentional herding and spurious herding. Intentional herding occurs when investors deliberately follow the actions of others, whereas spurious herding arises from independently correlated actions influenced by common external factors rather than direct imitation.

The Herding Index, used in this study to quantify herding behavior, is based on order imbalance, an indirect indicator of herding. However, order imbalance may result from various factors, including information asymmetry, market sentiment shifts, or genuine herding behavior. Although herding can contribute to order imbalance, not all instances of order imbalance necessarily indicate herding.

Due to the limitations of the available BSE data, distinguishing between intentional and spurious herding is currently infeasible. As a result, the SGIC model adopts a unified approach to measuring herding behavior. Future research can refine this model by incorporating agent tagging mechanisms, which would allow for the identification of agents who deliberately modify their decisions based on the actions of others. This enhancement would enable a more precise differentiation between intentional and spurious herding, thus improving the interpretability and robustness of the simulation results.

Additionally, future work could explore the following.

1. Incorporating heterogeneous investor types to better reflect real-world market diversity.
2. Testing the model against diverse financial datasets to enhance generalizability and predictive accuracy. Although the testing was done on highly volatile BSE data, the parameters considered have applicability in developed economy markets even with higher regulatory oversight and institutional investors. The behavioral mechanisms of herding, information cascade and SOC have been observed in major indexes like S&P 500. Agent-based models used here can be fine-tuned to reflect higher liquidity or even algorithm trading prevalent in developed markets.
3. Future studies will investigate the sensitivity of herding strength $|H|$ to the herding threshold θ , by varying θ across a range (e.g., 40%, 50%, 60%, 70%) and analyzing corresponding changes in order imbalance, even under stable avalanche statistics. In addition, a parametric sweep of the activation threshold $I_{th} \in \{0.5, 1.0, 1.5, 2.0\}$ and information dissipation rate $\beta \in \{0.50, 0.75, 0.98\}$ will be conducted to assess their impact on the power-law exponent α , the mean herding strength $|H|$, and volatility autocorrelation ρ_v . This exploration will help establish whether SOC dynamics and order-imbalance patterns persist across a broad spectrum of behavioral parameters, enhancing the generalizability and robustness of the SGIC model.
4. While the observed correlation between the imbalance measure H and preceding market activity highlights its potential as a systemic stress indicator, its utility as a forward-looking signal remains unverified. Future work will focus on the predictive assessment of H through rigorous methods such as Granger causality tests and Receiver Operating Characteristic (ROC) analysis. This will help determine whether H can reliably function as an early-warning signal for significant market fluctuations.
5. Future investigations will focus on the sensitivity of the SGIC model to memory length M and strategy reset threshold C . These parameters are central to agent adaptation and may significantly influence the emergence of self-organized criticality. Additionally, introducing strategic heterogeneity — by allowing agents to differ in memory span, update rules, or reset behavior — will enable a richer analysis of how agent diversity interacts with critical market states.

By building upon the strengths of the SG model and incorporating behavioral elements, the SGIC model aims to provide a more realistic representation of market dynamics and contribute to the advancement of financial modeling research.

Table 6
 ρ between herding index and index variations.

Window size	Preceding window	Succeeding window
2	0.2238	-0.0247
3	0.2671	-0.0288
4	0.2972	-0.0279
5	0.3249	-0.0256
6	0.3402	-0.0224
7	0.3544	-0.0270
8	0.3675	-0.0310
9	0.3767	-0.0312
10	0.3862	-0.0366
11	0.3911	-0.0410
12	0.3962	-0.0457
13	0.3992	-0.0461
14	0.4008	-0.0451
15	0.4022	-0.0437
16	0.4036	-0.0426
17	0.4032	-0.0422
18	0.4025	-0.0404
19	0.4007	-0.0408
20	0.3961	-0.0421
21	0.3924	-0.0404
22	0.3928	-0.0366
23	0.3906	-0.0342
24	0.3898	-0.0310
25	0.3893	-0.0287
26	0.3846	-0.0294
27	0.3833	-0.0277

(continued on next page)

Table 6 (continued).

Window size	Preceding window	Succeeding window
28	0.3815	−0.0256
29	0.3784	−0.0254
30	0.3761	−0.0240
31	0.3770	−0.0236

Table 7

Nomenclature.

Symbol	Description
α	Power-law exponent for avalanche distribution
β	Information dissipation rate
θ	Herding threshold (percentage of neighboring agents needed to follow herd)
ρ_v	Volatility autocorrelation
B	Board lot amount
C	Cognitive threshold
H	Herding index measure: $\frac{\text{Total Buy} - \text{Total Sale}}{\text{Total Buy} + \text{Total Sale}}$
I_{th}	Activation threshold for information transmission
K	Number of neighbors each agent interacts with in the network topology; governs connectivity and information flow
S	Number of strategies

CRedit authorship contribution statement

Sawar Sagwal: Conceptualization, Simulation, Literature Review & Writing and Editing. **Parthajit Kayal:** Conceptualization, Writing – review & editing. **Kavita Vemuri:** Conceptualization, Literature Review & Writing and Editing.

Appendix

See Tables 6 and 7.

Data availability

Data will be made available on request.

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